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Object Oriented Programming (Java)

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Outline

- What Is Java?
- Goal of Java
- Java Features
- Java as an OOP language
- Java Edition
- Installation Requirements
- Setting Path and Class Path Environment
- Creating a batch file for Command
- First Program
- Running a Java Application
- Typical Java Environment

What Is Java?

☐ "Java" generally refers to a combination of three things:

- The Java programming language
 - a high-level, object-oriented programming language
- The Java Virtual Machine
 - a high-performance virtual machine that executes bytecodes on a specific computing platform, typically abbreviated JVM
- The Java Platform
 - a JVM running compiled Java bytecodes, usually calling on a set of standard libraries such as those provided by Java Standard Edition (SE) or Enterprise Edition (EE).

Goal of Java

A key goal of Java is to be able to write programs that will run on a great variety of computer systems and computer-control devices. This is sometimes called “write once, run anywhere”.

Java Features

☐ Java has the following features:

- Platform Independent
- Simple
- Object Oriented
- Robust
- Distributed
- Portable
- Dynamic
- High-Performance
- Multi-threaded
- Interpreted
- Architecture-neutral

Java Features (Cont...)

❑ "Java" features in detail:

- **Platform Independent**

The concept of Write-once-run-anywhere (known as the Platform independent) is one of the important key feature of java language that makes java as the most powerful language.

- **Simple**

There are various features that makes the java as a simple language. Programs are easy to write and debug because java does not use the pointers explicitly. Java provides the bug free system due to the strong memory management. It also has the automatic memory allocation and deallocation system.

- **Object Oriented**

Java is a fully Object Oriented language because object is at the outer most level of data structure in java. No stand alone methods, constants, and variables are there in java. Everything in java is object even the primitive data types can also be converted into object by using the wrapper class.

Java Features (Cont...)

- **Robust**

Java provides the powerful exception handling and type checking mechanism as compare to other programming languages. Compiler checks the program whether there any error and interpreter checks any run time error and makes the system secure from crash. All of the above features makes the java language robust.

- **Distributed**

The widely used protocols like HTTP and FTP are developed in java. Internet programmers can call functions on these protocols and can get access the files from any remote machine on the internet rather than writing codes on their local system.

- **Portable**

The feature Write-once-run-anywhere makes the java language portable provided that the system must have interpreter for the JVM. Java also have the standard data size irrespective of operating system or the processor. These features make the java as a portable language.

Java Features (Cont...)

- **Dynamic**

While executing the java program the user can get the required files dynamically from a local drive or from a computer thousands of miles away from the user just by connecting with the Internet.

- **Performance**

Java uses native code usage, and lightweight process called threads. In the beginning interpretation of bytecode resulted the performance slow but the advance version of JVM uses the adaptive and just in time compilation technique that improves the performance.

Java Features (Cont...)

- **Multi-threaded**

Java is also a Multithreaded programming language. Multithreading means a single program having different threads executing independently at the same time. Multiple threads execute instructions according to the program code in a process or a program. Multithreading works the similar way as multiple processes run on one computer.

- **Interpreted**

With an interpreted language such as Java, programs run directly from the source code. The interpreter program reads the source code and translates it on the fly into computations.

Java Features (Cont...)

- **Architecture Neutral**

Java is an architectural neutral language as well. The growing popularity of networks makes developers think distributed. In the world of network it is essential that the applications must be able to migrate easily to different computer systems. Not only to computer systems but to a wide variety of hardware architecture and Operating system architectures as well. The Java compiler does this by generating byte code instructions, to be easily interpreted on any machine and to be easily translated into native machine code on the fly. The compiler generates an architecture-neutral object file format to enable a Java application to execute anywhere on the network and then the compiled code is executed on many processors, given the presence of the Java runtime system.

Java as an OOP language

❑ To be an Object Oriented language, any language must follow at least the four characteristics.

- **Inheritance**

It is the process of creating the new classes and using the behavior of the existing classes by extending them just to reuse the existing code and adding the additional features as needed.

- **Encapsulation**

It is the mechanism of combining the information and providing the abstraction.

- **Polymorphism**

As the name suggest one name multiple form, Polymorphism is the way of providing the different functionality by the functions having the same name based on the signatures of the methods.

- **Dynamic Binding**

Sometimes we don't have the knowledge of objects about their specific types while writing our code. It is the way of providing the maximum functionality to a program about the specific type at runtime.

Java Edition



- **Java SE**

Core Java Language. Desktop application.

- **Java EE**

Developing large-scale, distributed networking applications and web-based applications

- **Java ME**

Developing applications for small, memory-constrained devices, such as BlackBerry smartphones

Installation Requirements



- **JDK**

Java SE Software Development Kit (JDK)

- **Editor**

Popular IDE are

NotePad

Eclipse

NetBeans

jGRASP

JCreator

DrJava

BlueJ IDE

TextPad

Setting Path and Class Path Environment

❑ **Setting the PATH Environment Variable**

set path=C:\Program Files (x86)\Java\jdk1.7.0_25\bin

❑ **Setting the CLASSPATH Environment Variable**

set classpath=.;C:\Program Files (x86)\Java\jdk1.7.0_25\bin;
C:\Program Files (x86)\Java\jdk1.7.0_25\lib

Creating a batch file for Command

```
set path=C:\Program Files (x86)\Java\jdk1.7.0_25\bin
```

```
set classpath=.;C:\Program Files (x86)\Java\jdk1.7.0_25\bin;C:\Program  
Files (x86)\Java\jdk1.7.0_25\lib
```

```
c:\windows\system32\cmd.exe
```

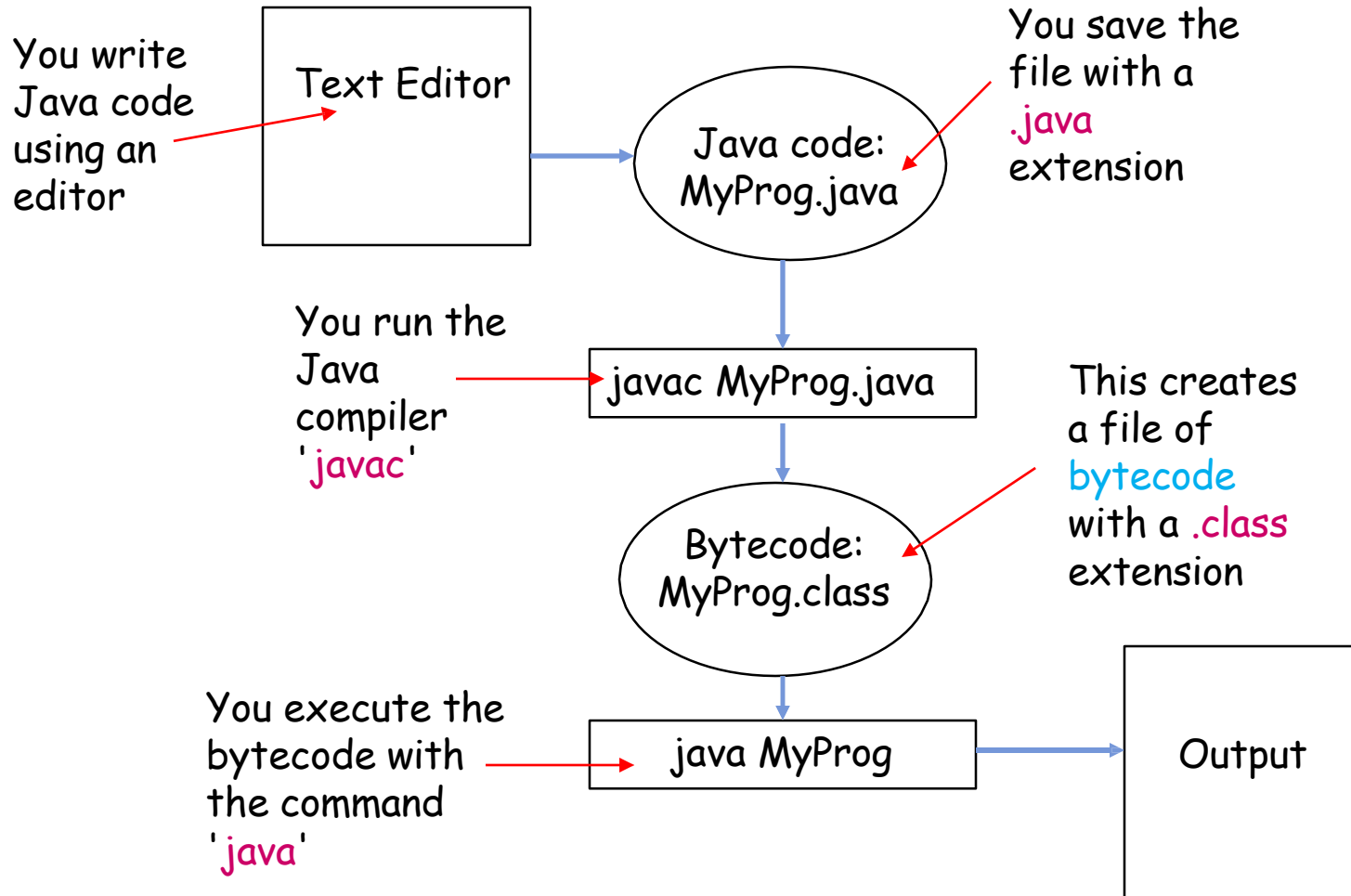
First Program

```
public class HelloWorld {  
    public static void main(String args[]) {  
        System.out.println("Hello World");  
    }  
}
```

The main() method is the starting point of every Java application. It looks like this because

- main() must be declared as public, since it must be called by code outside of its class when the program is started
- void indicates that this method will not return any information
- static allows main() to be called without having to instantiate a particular instance of the class
- String args[] declares a parameter named args, which is an array of instances of the class String.

Running a Java Application



Typical Java Environment

❑ Java programs normally undergo five phases

- **Edit**

Programmer writes program (and stores program on disk)

- **Compile**

Compiler creates bytecodes from program

- **Load**

Class loader stores bytecodes in memory

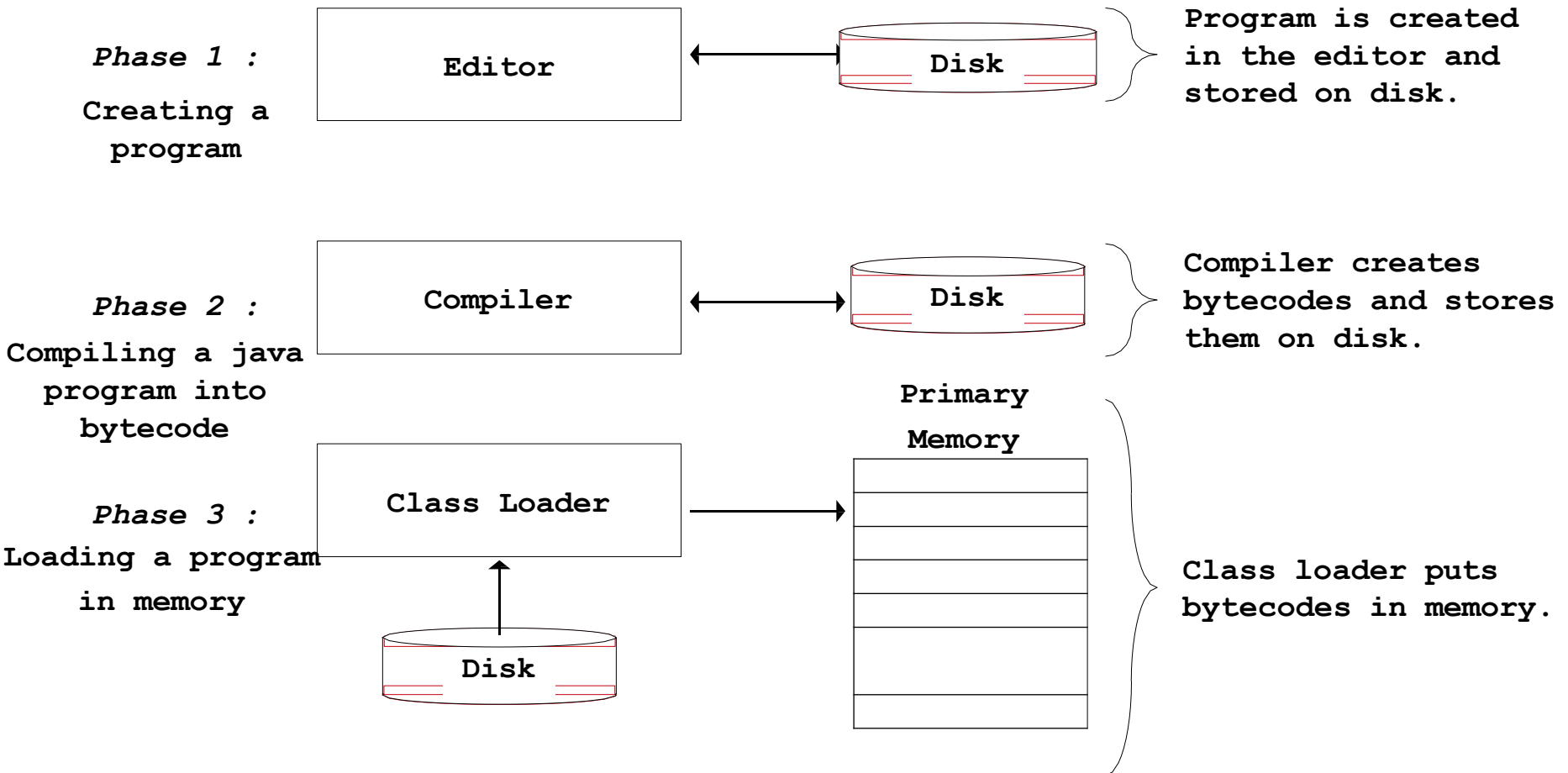
- **Verify**

Verifier ensures bytecodes do not violate security requirements

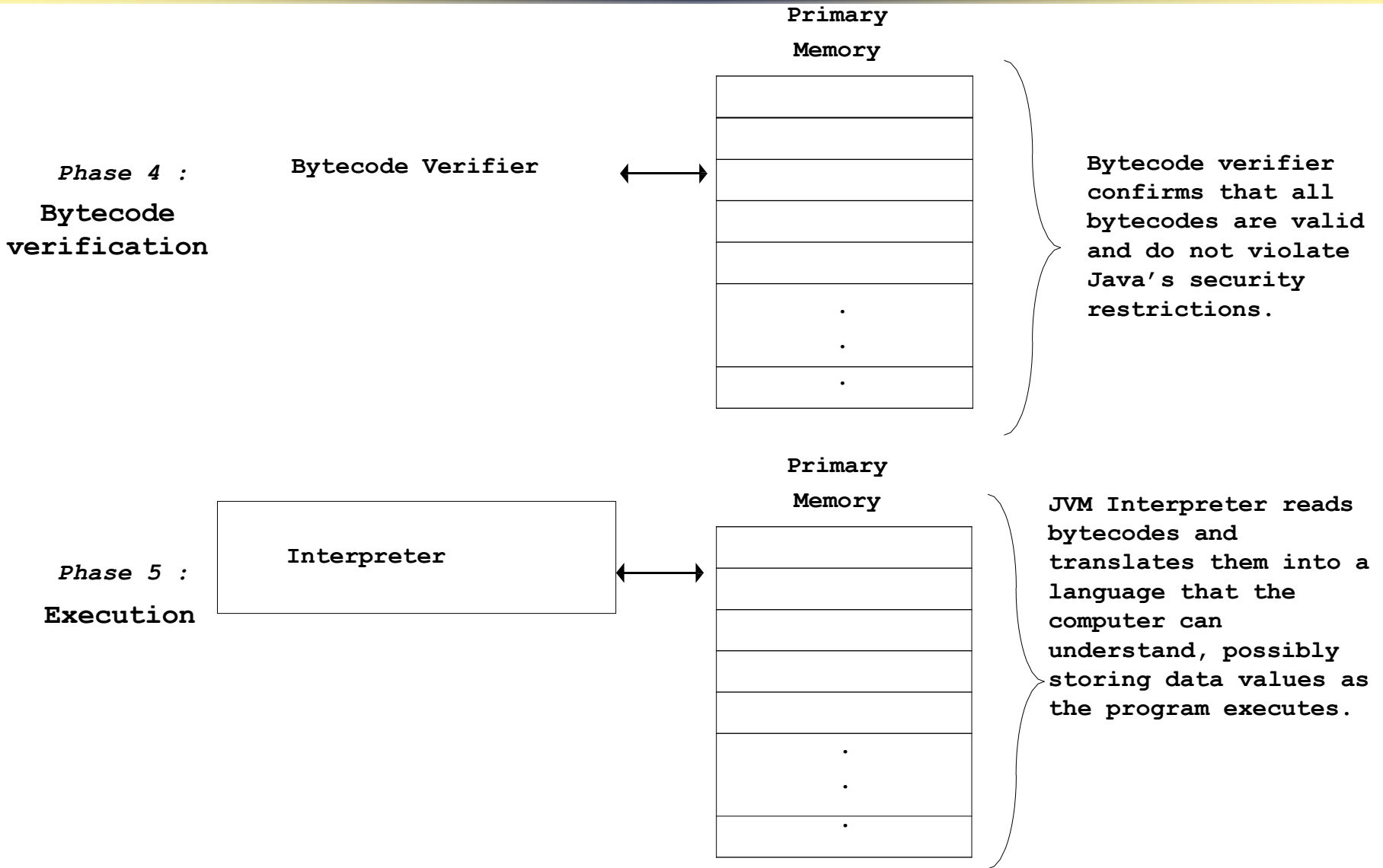
- **Execute**

Interpreter translates bytecodes into machine language

Typical Java Environment



Typical Java Environment



Books Recommended

1. Herbert Schildt : Java™: The Complete Reference, Ninth Edition
2. H.M. Deitel and P.J. Deitel : Java™: How to Program, Ninth Edition

Enough For Today

Thanks all of you.